

Recover deleted or damaged files from a storage device using TestDisk

EXPERIMENT NO: 02

DATE:

1. AIM

To recover deleted files and repair a missing or corrupted partition from a storage device using the TestDisk data recovery command-line utility.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tool:** TestDisk (v7.3-WIP Data Recovery Utility)
- **Hardware:** Target Storage Device (e.g., [\\.\PHYSICALDRIVE1](#) - USB SanDisk [3.2Gen1](#) USB device[30GB USB]).

3. DESCRIPTION

TestDisk is a powerful, open-source data recovery software primarily designed to help recover lost partitions and make non-booting disks bootable again when these symptoms are caused by faulty software, certain types of viruses, or human error (such as accidentally deleting a Partition Table). It can also be utilized to undelete files from FAT, exFAT, NTFS, and ext2 file systems. TestDisk queries the BIOS or the OS in order to find the hard drives and their characteristics, performs a quick check of the disk structure, and looks for lost partitions or deleted files.

4. PROCEDURE

Step 1: Log Creation & Disk Selection

1. Open the testdisk_win.exe application as an Administrator.
2. **Log Creation:** Choose [Create] to instruct TestDisk to create a log file containing technical information and messages. Press Enter to proceed.

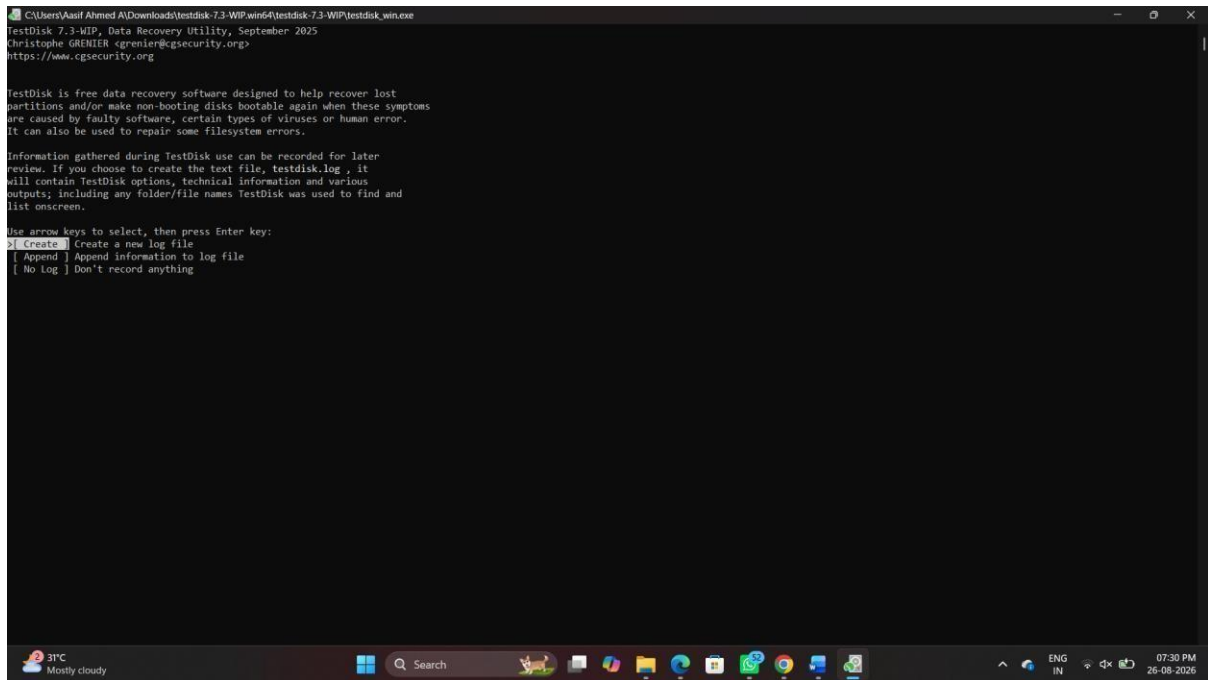


Figure 1: Initial TestDisk dashboard prompting for log creation

3. **Disk Selection:** All connected hard drives will be detected and listed. Use the up/down arrow keys to select the target hard drive containing the lost partitions or deleted files (e.g., \\.\PhysicalDrive1 - USB SanDisk 3.2Gen1 USB device[30GB USB]). Press Enter to Proceed.

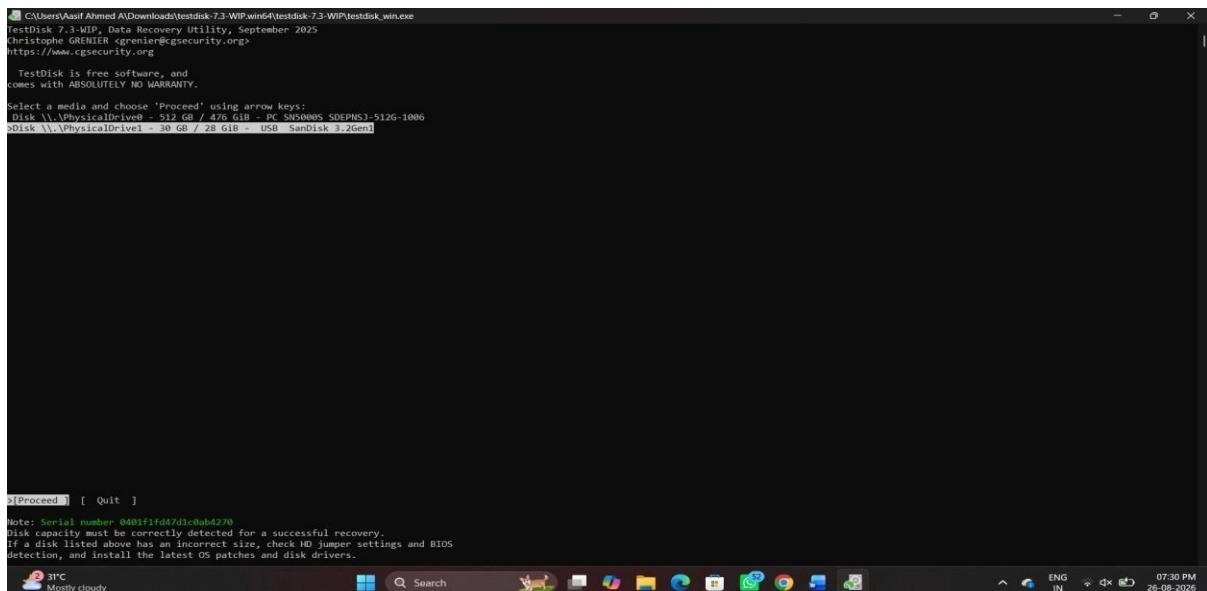


Figure 2: Selecting the physical storage drive in TestDisk Step

2: Partition Table Type Selection

1. TestDisk displays the available partition table types.

2. Select the partition table type. Usually, the default value is the correct one as TestDisk auto-detects the partition table type (e.g., [Intel] Intel/PC partition). Press Enter to Proceed.

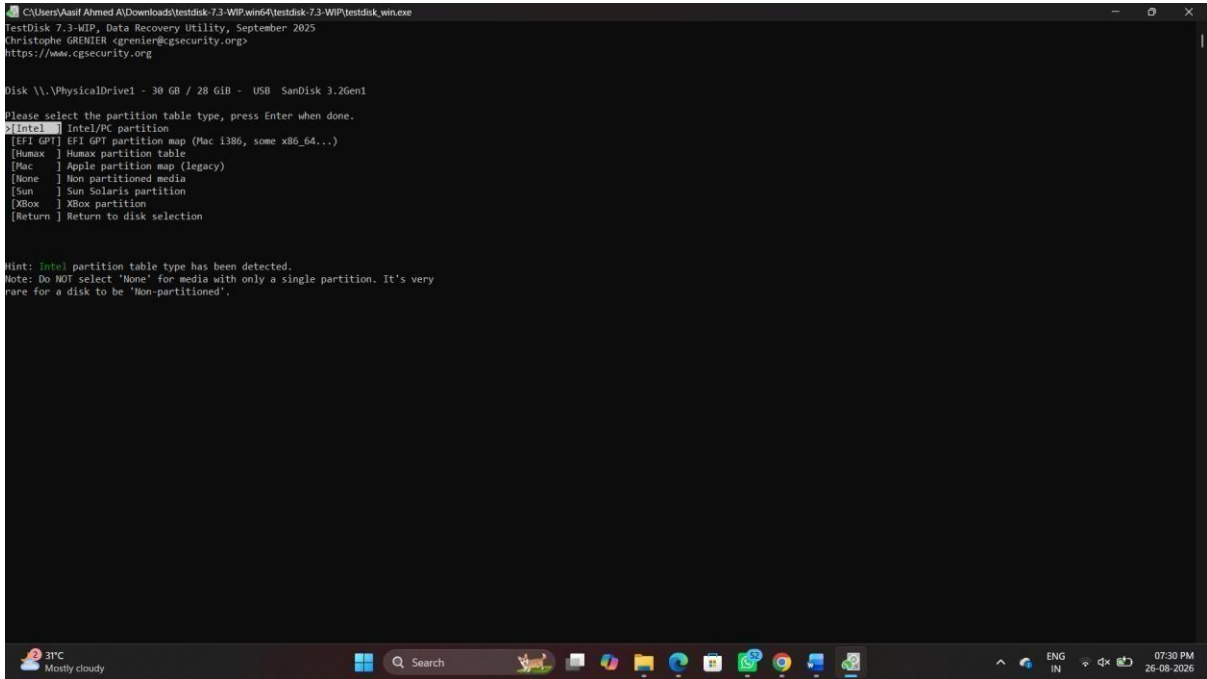


Figure 3: Selecting the Intel/PC partition table type

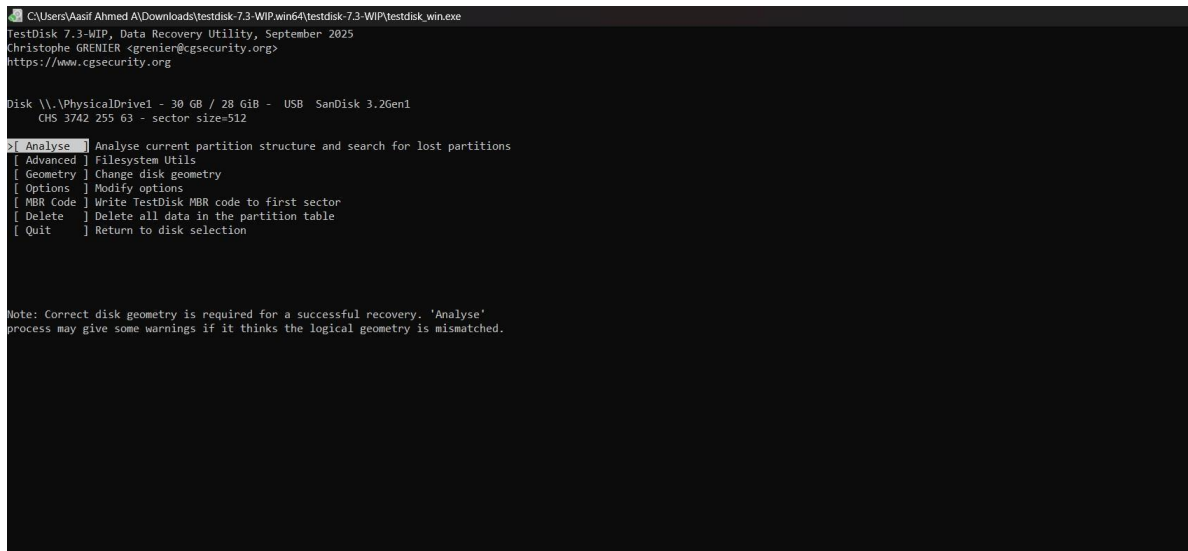


Figure 4: Choosing the Analyse option to examine the partition structure

Step 3: Current Partition Table Status

1. TestDisk displays the main menu options. Use the default menu [Analyse] to check your current partition structure and search for lost partitions. Confirm at Analyse with Enter.
2. The current partition structure is listed. Examine the structure for missing partitions and errors (e.g., Invalid NTFS boot, missing logical partitions). Confirm at [Quick Search] to proceed.

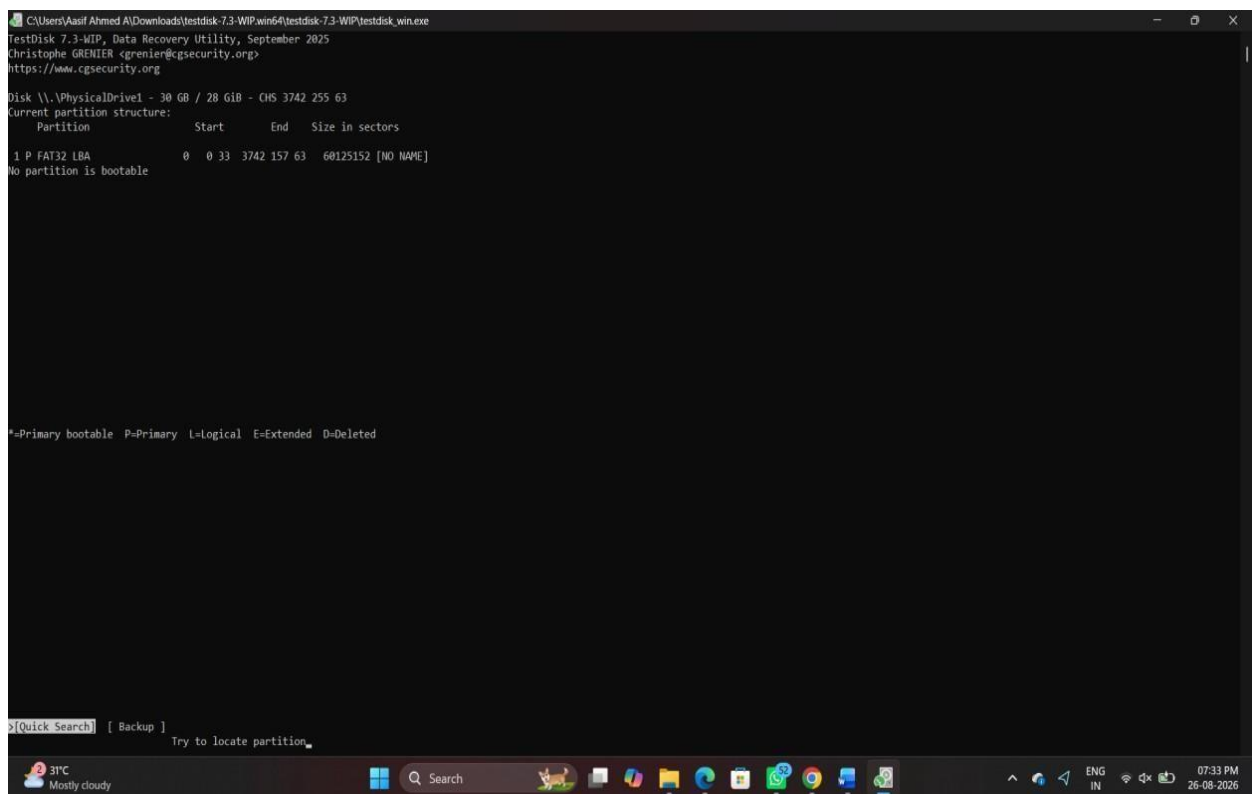


Figure 5: Initiating a Quick Search for lost partitions

Step 4: Identifying and Accessing the Correct Partition

1. Once the search is complete, highlight the correct partition you wish to recover or extract files from (e.g., >L FAT32 LBA).
2. Press the p key to list the files inside that partition.

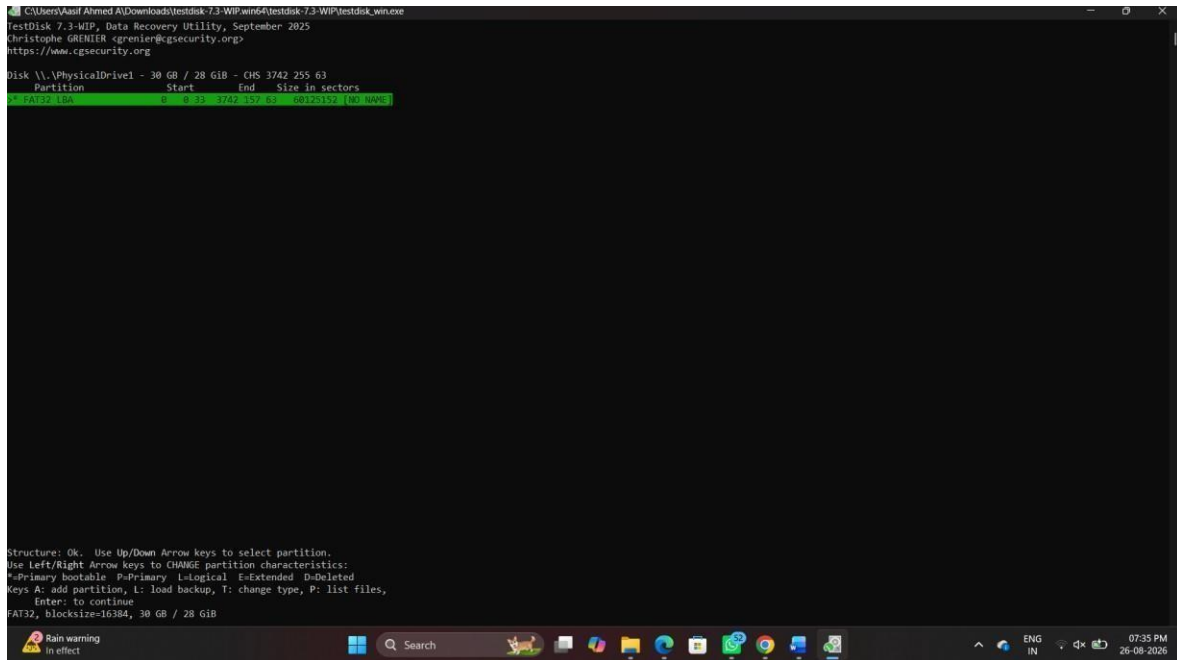


Figure 6: Highlighting the located FAT32 partition

Step 5: Recovering / Copying Deleted Files

1. In the file listing, navigate through the folders using arrow keys. Files listed in red indicate deleted entries.
2. Highlight the specific deleted file you want to recover (e.g., images (1).jpg).

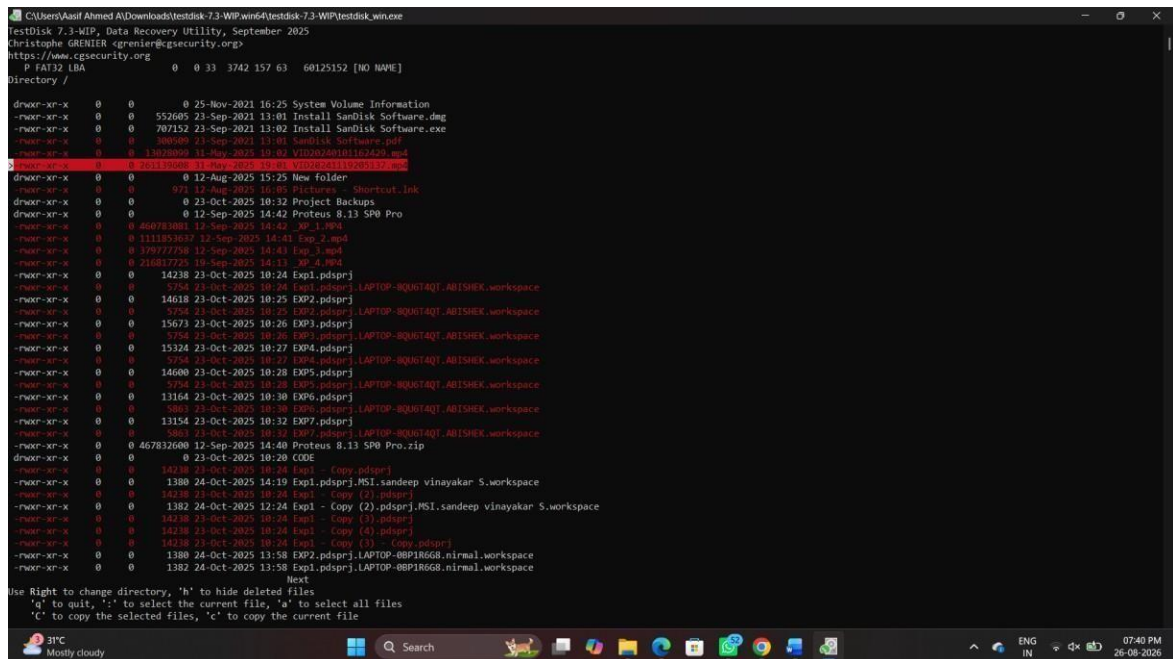


Figure 7: Listing the directory contents and selecting a file for recovery

3. Press **c** to copy the current selected file (or **C** to copy all selected files).
4. You will be prompted to select a destination directory to save the recovered file. Navigate to the desired local folder and press **C** when the destination is correct.

Figure 8: Selecting the destination folder for the recovered file

- ```
C:\Users\Asif Ahmed\Downloads>testdisk-7.3-WP-win64\testdisk-7.3-WP\testdisk_win.exe
testdisk 7.3 WP: Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
P FAT32 LBA 0 0 33 3742 157 63 60125152 [NO NAME]
Directory \VID020401119205137.sp4
Copy done! 1 ok, 0 failed
drwx--x--x 0 0 0 25-Nov-2021 16:25 System Volume Information
drwx--x--x 0 0 0 552605 23-Sep-2021 13:01 Install SanDisk Software.dmg
-rwx--x--x 0 0 0 707152 23-Sep-2021 13:02 Install SanDisk Software.exe
-rwx--x--x 0 0 0 500500 23-Sep-2021 13:01 Install Software.pdf
-rwx--x--x 0 0 0 13028099 31-May-2025 19:02 VID0204010107420.sp4
-rwx--x--x 0 0 0 25111096 12-Sep-2025 14:41 Exp1.pdspd
drwx--x--x 0 0 0 12-Aug-2025 15:25 New folder
-rwx--x--x 0 0 0 971 12-Aug-2025 16:05 Pictures - Shortcut.lnk
drwx--x--x 0 0 0 23-Oct-2025 10:32 Project Backups
-rwx--x--x 0 0 0 12-Sep-2025 14:42 Proteus 8.13 SP0 Pro
-rwx--x--x 0 0 0 460783061 12-Sep-2025 14:42 Exp 2.sp4
-rwx--x--x 0 0 0 1111853637 12-Sep-2025 14:41 Exp 2.mpd
-rwx--x--x 0 0 0 37977750 12-Sep-2025 14:43 Exp 3.sp4
-rwx--x--x 0 0 0 216819705 10-Sep-2025 14:11 Exp 4.mpd
-rwx--x--x 0 0 0 14238 23-Oct-2025 10:24 Exp1.pdspd
-rwx--x--x 0 0 0 5754 23-Oct-2025 10:24 Exp1.pdspd.LAPTOP-BQ6MTAQ7-AB15HEK.workspace
-rwx--x--x 0 0 0 14618 23-Oct-2025 10:25 EXP2.pdspd
-rwx--x--x 0 0 0 5754 23-Oct-2025 10:25 EXP2.pdspd.LAPTOP-BQ6MTAQ7-AB15HEK.workspace
-rwx--x--x 0 0 0 15673 23-Oct-2025 10:26 EXP3.pdspd
-rwx--x--x 0 0 0 5754 23-Oct-2025 10:26 EXP3.pdspd.LAPTOP-BQ6MTAQ7-AB15HEK.workspace
-rwx--x--x 0 0 0 15324 23-Oct-2025 10:27 EXP4.pdspd
-rwx--x--x 0 0 0 5754 23-Oct-2025 10:27 EXP4.pdspd.LAPTOP-BQ6MTAQ7-AB15HEK.workspace
-rwx--x--x 0 0 0 14600 23-Oct-2025 10:28 EXP5.pdspd
-rwx--x--x 0 0 0 5754 23-Oct-2025 10:28 EXP5.pdspd.LAPTOP-BQ6MTAQ7-AB15HEK.workspace
-rwx--x--x 0 0 0 13164 23-Oct-2025 10:30 EXP6.pdspd
-rwx--x--x 0 0 0 586 23-Oct-2025 10:30 EXP6.pdspd.LAPTOP-BQ6MTAQ7-AB15HEK.workspace
-rwx--x--x 0 0 0 13154 23-Oct-2025 10:32 EXP7.pdspd
-rwx--x--x 0 0 0 5863 23-Oct-2025 10:32 EXP7.pdspd.LAPTOP-BQ6MTAQ7-AB15HEK.workspace
-rwx--x--x 0 0 0 467812660 12-Sep-2025 14:50 Proteus 8.13 SP0 Pro.zip
drwx--x--x 0 0 0 0 23-Oct-2025 10:20 CODE
-rwx--x--x 0 0 0 14238 23-Oct-2025 10:24 Exp1 - Copy.pdspd
-rwx--x--x 0 0 0 1380 24-Oct-2025 14:19 Exp1.pdspd.JSI.sandeep vinayakar S.workspace
-rwx--x--x 0 0 0 14238 23-Oct-2025 10:24 Exp1 - Copy (2).pdspd
-rwx--x--x 0 0 0 1382 24-Oct-2025 12:24 Exp1 - Copy (2).pdspd.JSI.sandeep vinayakar S.workspace
-rwx--x--x 0 0 0 14238 23-Oct-2025 10:24 Exp1 - Copy (3).pdspd
-rwx--x--x 0 0 0 14238 23-Oct-2025 10:24 Exp1 - Copy (4).pdspd
-rwx--x--x 0 0 0 14238 23-Oct-2025 10:24 Exp1 - Copy (5).pdspd
-rwx--x--x 0 0 0 1380 24-Oct-2025 13:58 EXP2.pdspd.LAPTOP-0BP186G8.nirmal.workspace
-rwx--x--x 0 0 0 1382 24-Oct-2025 13:58 Exp1.pdspd.LAPTOP-0BP186G8.nirmal.workspace
Next
Use Right to change directory, 'h' to hide deleted files
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file
```

### Step 6: Partition Table and Boot Sector Recovery (Optional)

1. Press q to quit the file listing and return to the partition selection.
2. Use the left/right arrow keys to change the status of a partition from D (Deleted) to L (Logical) or P (Primary) to recover the partition itself. Press Enter.
3. If all partitions are correctly listed, confirm at [Write] with Enter, press y to confirm, and then OK.
4. If a boot sector is damaged, use the [Backup BS] option to copy the backup boot sector over the damaged one.
5. Exit TestDisk and reboot the computer to access the restored partition data.

### RUBRICS:

| Criteria & Marks Assigned                              | Mark Allotted | Mark Awarded |
|--------------------------------------------------------|---------------|--------------|
| 1. GitHub Activity & Submission Regularity             | 3             |              |
| 2. Application of Forensic Tools & Practical Execution | 3             |              |
| 3. Documentation & Reporting                           | 2             |              |
| 4. Engagement, Problem-Solving & Team Collaboration    | 2             |              |
| Total                                                  | 10            |              |

### 5. RESULT

The TestDisk data recovery utility was successfully executed to analyze the physical drive's geometry and partition tables. Lost partitions were successfully identified through deeper searching, and deleted files were effectively navigated, selected, and copied (recovered) to a secure local destination.